

M002 · AND Conditions

Topic: AND — two things both true · **Course:** Maze Madness · **Level:** Intermediate · **Time:** about 35 minutes

AND checks two things. The action runs only when **both** are true.

1 · Predict 🧠

```
if wall ahead AND wall on right:
    turn left
```

Both true. The agent does what?


```
if no wall ahead AND no wall below:
    move forward
```

Path ahead is clear, but a wall is below. Does it move? Why?

2 · Trace

```
keep doing forever:
  if wall ahead AND wall on right:
    turn left
  otherwise:
    move forward
```

At a corner (wall ahead AND wall on right), what happens?

At an open path, what happens?

3 · Fill the Code

Turn left only when both walls are there.

```
if agent.detect(BLOCK, FORWARD) ____ agent.detect(BLOCK, RIGHT):  
    agent.turn(____)
```

4 · Spot the Bug 🐛

Should turn left only at a corner: wall ahead AND wall on right.

```
if agent.detect(BLOCK, FORWARD):  
    agent.turn(LEFT_TURN)
```

Write the fix:

Why was it wrong?

5 · Show Your Work 📷 🎥

Solve the maze with one **AND** .

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.